

What is a theory and what constitutes a theoretical explanation?

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1.0. Introduction

The purpose of this text is to explain the concept of a theory, and what theories are good for. More specifically, I will address the following topics: a) meaning of the word theory in everyday life vs. research, b) two different perspectives on what constitutes a theoretical explanation, c) difference between a hypothesis and a theory, d) how inductive & deductive reasoning support theory building & testing, e) why it can be difficult to tell induction and deduction apart, and what abduction is. Overall, this provides an understanding of what a theory and a theoretical explanation actually is. Btw., terminologically speaking, the purpose of a theory is to explain something, so if you provide a theory, you have also provided an explanation.

It is important to note that the concept of a theory is complicated and can be approached in different ways. Such differences are due to different fundamental assumptions, different practices and focus points within different fields (e.g. economics vs. management), and the fact that it is simply difficult to pin down an abstract concept such as a theory (Hullman 2022). In this text, I will present a simplified and general overview of what a theory is, but it is important to keep in mind that more nuanced and different definitions and perspectives exist – some you might encounter in future studies at a master level. I will try to make disagreements and ambiguities explicit while focusing on the core elements that most agree on. I

also provide additional references and links, but you are not expected to read these for this course. They may be useful for a bachelor project or other purposes.

1.1. Theory in everyday life

What does the word theory mean? This is not an easy question, since the term is used in many different contexts, ranging from everyday life to sophisticated theoretical fields such as physics or economics. The term is thus ambiguous, and the meaning depends on the usage being referred to. In everyday life, we all implicitly or explicitly engage in theorizing on a regular basis (Gill & Johnson 2010, Zimmer 2016). In fact, *"In everyday conversation, we tend to use the word "theory" to mean a hunch, an idle speculation, or a crackpot notion. That's not what "theory" means to scientists."* (Zimmer 2016, p: 1). For example, someone might say 'I have a theory about why Peter is hesitant to meet up with us', or 'I have a theory about why we can't keep our new customers'. Such (implicit) theorizing has been part of human life for millennia; we learned that cooking water is better for one's survival, or that a given animal should be hunted in a certain way (Christakis 2019). Theories in everyday life are often not based on systematic scientific investigations, but rather based on personal experiences, common knowledge or stereotypes we adhere to (Gill & Johnson 2010). Irrespective of their origin, **theories are like maps that help us navigate the world** by simplifying the landscape they represent and telling us what we can expect to find in a given place (Zimmer 2016, Toulmin 1953).

Note that if you have no theories, it is difficult to make sense of data, since one needs an overarching framework and helpful concepts to organize data and tell you what relations one can expect to find in data. Otherwise data are just data points with no labels and no explanations. For example, if you want to understand the current GDP growth of a country, you need to use concepts and terminology such as GDP, economic growth, and economic indicators, which all originate from theoretical frameworks in economics. We can't measure the world without some kind of (theoretical) concepts being used. Even just plotting data on a graph involves theory; one has decided to categorize the x and y axis in a certain way. Therefore, when it comes to complicated business administration and economics questions, there is no such thing as a 'pure non-theoretical dataset', since they all in the end rely on definitions and frameworks. In other words, this is why supervisors always expect you to use theory in your bachelor / master projects. Similarly, in research and everyday life theories serve as a helpful way of simplifying a complex (business) world. If something unexpected happens, we might even change our (theoretical) expectations (Gill & Johnson 2010: p. 44-45), although such changes in our worldviews probably happen less often than it should (Kahneman 2011, Galef 2022, Kuhn 1970).

Being better at building, testing and using theories is not just for academics working with scientific data, but also for individuals in their everyday life and individuals in organizations wanting to improve organizational performance. The basic idea is that the *better theories we work with, implicitly or explicitly, the better we can manage the world.*

1.2. What researchers do: Description vs. Explanation vs. Prediction

Einstein stated: *"[t]he whole of science is nothing more than a refinement of everyday thinking."* (cited in Mantzavinos 2019: p. 778). In this perspective, science and scientific thinking is not something only experts engage in. Science is not fundamentally distinct from everyday life. Yet: *"In science, the word theory isn't applied lightly," Kenneth R. Miller, a cell biologist at Brown University, said. "It doesn't mean a hunch or a guess. A theory is a system of explanations that ties together a whole bunch of facts..."* (Zimmer 2016, p: 1).

In science, we engage in systematic work and for example do not just rely on anecdotal experiences. Yet, what does this mean and entail? In other words, what do researchers in business administration and economics do when they engage in scientific work, that sets them apart from everyday life theorizing?

In order to address that question, we first need to make two related distinctions; the first between **descriptions** and **explanations**. Scientists engage in a range of activities, as highlighted by Heckman (winner of the 'Nobel Prize' in economics in 2000). "*Empirical analyses in economics have diverse goals—all valuable. Some analyses advance knowledge by uncovering new facts or providing richer descriptions of old facts ... Other analyses seek to understand the [theoretical] mechanisms producing outcomes with an eye toward interpretation and counterfactual policy evaluation.*" (Heckman and Singer 2017: p. 298).

As the quote says, doing research implies a broader set of activities than merely building or testing theory. Sometimes we just want to uncover new facts and categories, and describe what is going on; what educational backgrounds do CEOs in various industries have, in what contexts are chatbots being used in organizations, what is the annual growth rates of stocks etc. Just knowing such descriptive numbers can be relevant. For example, maybe we realize that some types of educations are over-represented among CEOs, compared to a baseline. That is if 10% of the population has X education, but 15% of CEOs have X education, they are more likely to be CEOs. Now, such descriptions do not explicitly entail any **explanations** of **why** this is the case. From a **descriptive** point of view, we might merely be interested in questions like who, what, when and where. Good descriptions can set the stage for a further investigations of explaining why something is the case.

The second distinction is between **predictions** and **explanations** (Barrowman 2014, Hofman et al. 2017). Large datasets might allow you to make precise and useful predictions about what will happen. For example, a large organization might have collected so much data about their customers, that they can predict which customers are more likely to buy a certain product. However, such a prediction does not necessarily imply knowing **why** there is such a relation - which is what we are focused on here; **theoretical explanations**. A key point is that knowing why there is a relation provides a substantial advantage, since if we don't why a relation exists, we will struggle to know if a relation will continue to hold when the context changes (Hofman et al. 2017).

Table 1: Illustrating the three concepts: Description, Explanation and Prediction

Concept	Definition	Example: Electric Vehicles in Europe
Description	Stating facts, figures, or characteristics about a phenomenon without explaining why it occurs. It's about the "what," "where," "when," and "how much."	The market share of electric vehicles in Europe was 25% in 2023.
Explanation	Providing reasons or underlying mechanisms that account for a phenomenon. It addresses the why question.	Consumers choose electric vehicles over gasoline-powered ones due to environmental concerns, lower running costs, government incentives, and improving technology (e.g., range, charging time).

Prediction	Forecasting future outcomes or behaviors. Ideally, it is based on an understanding of underlying causes (the "why"), allowing for more robust and generalizable forecasts. However, predictions can also be made based on observed data patterns (correlations), although these may be less reliable, especially in changing contexts or when spurious correlations exist.	A 10% reduction in government subsidies for electric vehicles is expected to decrease their sales by 5% in the next year. This prediction is more likely to be accurate if it's based on a good understanding of why subsidies influence EV sales. A purely data-driven prediction might not hold if other factors (e.g., battery technology) change significantly.
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In the following I expand upon what it means to say one has provided a theoretical explanation. Hereafter I will clarify that different scientific fields and communities can have fundamentally different expectations of what a theory can and should do.

1.3. What is a theoretical explanation?

"An [theoretical] explanation is an attempt to answer the question of why a particular phenomenon occurs or a situation obtains, that is, an attempt to provide understanding of the phenomenon or the situation by presenting a systematic line of reasoning that connects it with other accepted items of knowledge" (Henk W. de Regt, *Understanding Scientific Understanding*, 2020 [2017], Oxford University Press, pp. 24-25).

We focus here on theoretical explanations about empirical phenomena. The presentation below does not capture theory on theory or so-called grand theories such as Marxism, capitalism, evolution etc. (Merton 1968).

A theoretical explanation involves a *"...strategic reduction: abstracting away enough of the world's confusion to develop pointed explanations of organizational phenomena"* (DiMaggio 1995: p. 393). Why would you want to make the explanation more abstract? If you don't focus on general, abstracted features of organizational phenomena, you only explain one unique situation. That would only be of limited use. By 'abstracting away' you can generalize and broaden the explanation, making it useful for a broader audience. So, rather than only explaining why this particular CEO does this or that, we want to explain why CEOs (or this kind of CEO) generally does this or that. A theory can be used to explain one unique situation, but would be able to do so because general features of the phenomena can be recognized.

Yet, what does this **'strategic reduction'** and **'abstracting away'** really mean? Let us refer to the metaphor of a **theory as a map** of the world mentioned earlier. If a map of the world is as big as the world, it would be useless, so obviously any geographical map also 'abstracts away'. A good geographical map of the world highlights the elements one is interested in (Toulmin 1953), be it roads, elevations, cities, temperatures etc. The same world can thus be mapped in many ways. *Similarly, a good map of the world of business administration, highlights the elements we are interested in.* For example, if you are trying to develop a theory of effective leadership, you cannot simply provide a complete and detailed account of every action, decision, and interaction that successful leaders engage in. This would be impractical and would likely result in an explanation that is too long and complicated to be useful. Instead, you need to strategically reduce the problem and focus on the core issues that are most relevant to understanding effective leadership.

Note that a theoretical explanation should show the underlying (e.g. psychological, economic or sociological) drivers and dynamics of a given relation (Whetten 1989: p. 491). **In other words, the ‘why’ explains underlying mechanisms or processes that produce phenomena.** In Sutton & Staw’s (1995) terms, the answer to this question spells out the **underlying logic of the relation**. For example, just stating that higher incentives can lead to superior performance does not explain why this relation might exist. Maybe this seems obvious to some readers, but it is still necessary to explicitly spell the underlying logic out, if we want to provide a clear theoretical explanation.

Let’s build on the incentives and performance example: It could be that higher incentives drives the individual to put in a higher effort (see also Holst-Hansen & Bergenholtz 2020). The relation might therefore be driven by the individual’s willingness to invest more hours to achieve a higher performance. However, the higher performance could also be due to incentives driving individuals to strategically change their approach to how to carry out a given task. Or, the higher incentives could lead to a sense of (social) expectation and the individual might feel obliged to perform better. By explicitly spelling out the logic or reasoning behind a particular explanation, you can better communicate your understanding of the phenomenon and ensure that others in a different context might be able to transfer (i.e. generalize) your explanation into their situation. Importantly, all of the above explanations might have some truth to them, thus the explanations might be complementary. You should not think that only one explanation is the right one.

The former paragraphs presented a general angle on (and definition of) what it means to theoretically explain something. We can also try to be more specific and identify what elements a theory contains. Before that step, I want to note that Sutton & Staw (1995) actually begin from the opposite direction, by **highlighting what a theory is not**: If you present a theory, you should not only present a list of references, hypotheses, data and numbers, and graphs, since these elements do not constitute a theoretical explanation. This link provides a brief overview of their criteria of what a theory does not entail (https://faculty.babson.edu/krollag/org_site/craft_articles/sutton_theory_not.html).

2.0. Two approaches to theory: Causal explanation approach vs. Understanding explanations approach

The word ‘explanation’ is not necessarily straightforward. What does it, exactly, mean to have explained something and provided an answer to a ‘why’ question. Thus, what criteria and expectations should we have for a satisfying explanation? While several different approaches to these questions exist, this text relies on a basic distinction that other more nuanced terminologies and frameworks usually refer to. There are arguably two camps, focusing either on ‘**causality**’ or ‘**understanding**’, and they imply different perspectives on what we should expect from a theory. Briefly put, the ‘**causality**’ perspective argues that we should be concerned with identifying the underlying **mechanisms** or processes that produce a particular phenomenon and be able to **predict** the relationships between different factors or variables (Pearl and McKenzie 2018). These explanations often use analytical tools such as diagrams or statistical models to identify causal relationships and to make predictions about how different factors will affect outcomes. In contrast, following the perspective of ‘**understanding**’, one is focused on being able to understand and **make sense of a given situation**, in order to offer a rich and nuanced perspective. These kinds of explanations often focus on the **intentions** of human actions, in order to understand the meaning of them. This also implies a focus on contextual or qualitative factors and the approach may use qualitative methods such as interviews or case studies to provide a rich and nuanced understanding of the situation.

As indicated above, the fundamental differences are in some part related to the distinction between qualitative and quantitative work. However, do note that just because someone is engaged in qualitative (quantitative) work, does not per se mean that the aim to provide a causal (understanding) explanation. Eisenhardt (1989) e.g. makes clear that her - mainly qualitative - case study approach adopts a “positivist view of research” (p. 546), where the aim is to develop “testable hypotheses” (p. 546) and produce theory that “closely mirrors reality” (p. 547). She doesn’t use the word ‘cause’ in the text, yet it seems clear that causal relationships is what she is trying to establish – with qualitative work.

Furthermore, there are also general differences across disciplines. For example, economics generally expects researchers to be able to draw causal arrows between concepts and make (theoretical) predictions, while studies in organizational culture might focus on being able to understand the intentions and behaviors of individuals and organizations, rather than if consistent theoretical links between variables can be established (this might also be because studies have not been able to find a clear causal link between organizational cultures and performance). Thus, the latter argument is that even if we could create measures that allow statistical regression results to be reliably consistent and predictive, these numbers might not really tell us what is going on and how an individual should behave in any given context. Obviously, often both approaches are used and valued. In the following, an in-depth presentation of the two approaches will be provided.

2.1. Causal explanations

“If we cannot explain a phenomena in terms of underlying cause and effect, then we have not really explained anything” (Puranam 2018: p. 20).

A causal effect can be about a *single* event: Titanic hit an iceberg and it sunk. Similarly, a particular individual might have caused chaos in a project and made it fail. We are here interested in **general causation**. What might make ships sink or what traits, behaviors or processes might make projects fail? Therefore, in order to actually provide a theoretical explanation, we should provide a causal general explanation of what happened – either explaining the cause or effect of certain kinds of phenomena.

Note that providing a causal explanation does not necessarily require that we can provide final, causal *proof* of the relation, but the explanation should imply a causal relation and provide some evidence for it. For example, if I increase an incentive (the **cause**), performance will increase (the **effect**) due to an enhanced effort (the **why**). Hence, I would have both provided the ‘why’ of an explanation and I have shown how the cause (the incentive) is related to the effect (the performance). Often arrows would be drawn in a visual diagram where arrows indicate causal relationships between variables. A key challenge in such visualizations and one’s ability to understand causality is that causal drivers in business administration and economics are generally invisible. In physics causality can appear to be visible and immediate; in billiard you can see the first ball touch the other, where after the second ball immediately moves. Note that we don’t see causality as such, but see that something happens right away (Hume 1953). However, an incentive can redirect effort, aims or goals of an individual in a way, where the cause is distant from the effect, not visible at all and the effect could also be delayed.

Put differently, the idea is that we are not only able to describe what has happened, but what theoretically would happen, if we changed something – thus we think we know **the counterfactual** (as in a randomized controlled trial). Right now, an organization might provide X level of incentives, and changing that incentive

should have an impact. The perspective thus also, at least implicitly, confers some level of predictive control: I think I know what will happen, if I change this or that.

In the following I provide a few further reflections on key characteristics of this perspective on what an explanation is.

Probabilistic predictions – not universal laws

A theoretical prediction does not guarantee a deterministic cause and effect relation, where an effect always will follow from a cause. This type of relationship would be a universal law, like the law of gravity (see Nelson 2016, Watts 2014). In business administration, causal relationships are generally considered **probabilistic**, meaning that something is more or less likely to have an effect.

"Rather than asking whether a given theory can predict some outcome of interest, the accepted practice in social science instead asks whether a particular coefficient in an idealized model is statistically significant and in the direction predicted by the theory." (Hofman 2017, p: 486).

To illustrate, a theory *does not have to* predict and specify exactly how much performance will improve given certain incentives (e.g. by 13%). Instead, the aim is to show the direction of the expected performance change. For example, that a given cause (incentives) **on average** has a certain effect (performance). The 'on average' specification is often implicit, and not written explicitly. The reason researchers in these fields have to aim for probable and general relations is that business and economic contexts are complex, meaning that many different variables are potentially in play and could be changing at any point in time (Sullivan 2011, Watts 2007). Only in randomized controlled trials, not in everyday life, can we try to ensure that everything else is constant when investigating potential relations. I will expand on the nature of hypothesis testing below (section 3.1.1).

Finally, it is important to note that a prediction does not necessarily have to be about the future. It is possible to use historical data, such as events from 2001 to 2010, to build a theory and then test and predict events within that same dataset. For example, we could use half of the dataset (2001-2005) to build the theory and then apply it to the other half (2006-2010) to make predictions. This approach would allow the researcher to engage in theoretical prediction using historical data, rather than trying to predict future events.

Quantitatively specify the theoretical effect?

Nevertheless, even though that traditionally an explanation has merely been required to specify the direction of an effect (e.g. direction towards higher or lower performance), there is a more recent trend towards requiring more detailed specification of the expected strength of the effect. This means that in addition to indicating whether incentives will influence performance (in a binary 'yes' or 'no' sense), a theory should also provide an estimate of how large the effect is expected to be. It does make a substantial difference if the performance increase is going to be 1%, 10%, or 50%! If the theory then specifies that the effect should be at least 20%, and it turns out to be much smaller, the theory would have been shown not to be valid. Again, social situations consist of many different dynamic factors, so very precise predictions are generally not possible (unlike in physics, where we could exactly determine the impact something has). Yet, maybe we should expect more from our theoretical explanations, and at least be able to specify if the effect is expected to be small, medium or large (Hofman et al. 2017, Funder & Ozer 2019).

Qualitative approaches and causality

While the language involved in talking about causal explanation is usually more akin to the world of quantitative methods and data, a causal explanation can also be provided qualitatively (Watts 2017, Turco & Zuckerman 2017, Yin 2009). In this way, we could ethnographically observe behavior in a context, and be able to explain why some people behave as they do. Imagine that a company is trying to understand why its employees are not using a new productivity tool that has been introduced. The company could conduct an ethnographic study by observing employees in their work environment and conducting in-depth interviews with them. Through this research, the company may identify patterns in the way employees work and the reasons why they are not using the new tool. For example, this research may reveal that employees are not using the tool because it clashes with their incentive system or existing established methods for completing tasks. Note that such explanations do not have to take their own self-understanding ('I think this is the reason') at face value but could identify patterns that people might not even be aware of (see Turco & Zuckerman 2017 for examples of this).

2.2 Understanding Explanations (Verstehen): Seeing the World Through Others' Eyes

Unlike causal explanations that focus on identifying cause-and-effect relationships, the **understanding (Verstehen)** approach aims to make sense of the **subjective meanings** and experiences of individuals to explain their actions and behaviors. A researcher relying on an understanding perspective tries to see the world through the eyes of those researched, in order to understand the **intentions** of their actions, how they **interpret** the world, and the social **context** they are part of.

Beyond Cause-and-Effect

In other words, to explain is to reach a deeper understanding of social and business phenomena, from the human actors' point of view (Hofman et al. 2017, Watts 2014, Mantzavinos 2019). Simply identifying a relationship between variables is insufficient without understanding the human meaning behind it. For example, knowing that a competent leader influences followers is less useful without understanding what "competence" means to the followers, how they perceive their leader's actions, and how their interpretations affect their willingness to engage.

Causality in Verstehen

However, the understanding perspective is not a single, unitary position. It contains different views on the role of causality, with some researchers arguing that causality is secondary while others accept its primary relevance. Watts (2014: p. 318) states, "Although [**Verstehen**] explanations are often presented in the form of **causal explanations**—for example, they may claim to identify causal mechanisms and even speculate about counterfactuals—they are different in that they are **evaluated primarily, if not exclusively, in terms of their ability to make sense of some observed behavior or outcome**—that is, to render it understandable— by reducing it to intuitive, interpretable claims about human motives, incentives, and opportunities." (my italics). You might therefore both see texts or positions that rely on rich, qualitative insights and still offer a causal account (why something led to something else), while others would – explicitly or implicitly – argue that it is all about sensemaking, and we should not focus on offer causal links.

Overall, the understanding perspective recognizes that behaviors are influenced by individual interpretations and contexts. It does not necessarily reject causality but instead places primary importance on making sense of behavior through the lens of human motives and experiences. Thus, while **causal claims** can be made, they are **supplementary** rather than primary.

2.3 Comparing the two approaches

The two perspectives of causality and understanding can be distinguished not only by their focus but also by their methodological approaches and underlying philosophical assumptions.

Philosophical assumptions: The causality approach is often associated with positivism or post-positivism—focusing on objective data and measurable outcomes. In contrast, the understanding approach is more closely aligned with constructivism, which values the subjective constructions of reality as experienced by individuals, therefore emphasizing context and meaning over general explanations (cf. Guba 1990, curriculum in a later lesson). These differing philosophical assumptions shape how researchers frame their questions and interpret their finding (cf. Guba 1990, curriculum in a later lesson).

Data collection strategies: The **causality** approach typically favors **quantitative methods**, such as experiments and statistical models, as the preferred standard for demonstrating relationships and establishing generalizable findings. For instance, economists might use randomized controlled trials or analysis of large quantitative datasets to predict how changes in inflation rates influence consumer spending. This approach emphasizes the ability to generalize findings beyond specific contexts.

In contrast, researchers following the **understanding** approach often utilize **qualitative methods**, such as semi-structured interviews, participant observations, and case studies to collect data. For example, instead of merely predicting how changes in inflation rates influence consumer spending, an understanding approach might involve speaking directly to consumers to explore how they interpret these changes in the context of their everyday lives—how they think of the impact on their future financial plans, household security, and broader well-being. Data collection focuses on open-ended questions that allow participants to express their personal experiences freely, capturing their individual perspectives. Analysis of this data might involve thematic analysis to identify patterns and topics, as well as narrative analysis to understand the stories individuals tell about their financial behavior in response to interest rate changes.

Evaluating theory: The differences between the causality and understanding perspectives also lead to different assumptions and criteria for evaluating the quality of a theory. In the **causality** approach, the metaphor for validity is "**hitting the bullseye**." The quality of a theory is to be tested with the available data and assessed based on how closely it aligns with the truth or reality of the situation (even in a widely used qualitative approach, such as Eisenhardt (1989), see section 2.0). In contrast, the **understanding** approach does not rely on the concept of an independent bullseye as a measure of validity. Instead, it emphasizes **intersubjectivity**, **authenticity**, and **credibility** as core values and criteria for evaluating the quality of a theory (Guba & Lincoln 2001). While the understanding perspective may still use terms related to validity, it operates under different assumptions about what it means for a theory to be valid.

These two perspectives can sometimes complement each other. For instance, while a causal analysis might show that increased wages lead to higher productivity, an understanding approach might explore how workers interpret these wage increases—do they feel valued, motivated, or pressured? By examining these subjective elements, researchers can provide depth to the causal relationships and explain why certain outcomes occur under particular conditions.

Table 2: Using the case of Generative AI to illustrate differences

Generative AI: Balancing Prediction, Verstehen, and Explainability
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Introduction Generative AI (GenAI) is a powerful tool for making predictions, but it cannot (yet?) truly understand - in the sense of Verstehen. Verstehen focuses on grasping the meaning behind human actions, shaped by their intentions, experiences, and social context. While GenAI can imitate and simulate patterns in language and behavior, it does not interpret meaning as humans do.
GenAI and Its limits GenAI predicts outcomes by recognizing patterns in large datasets, but it operates as a “black box.” For example, it can predict which customers are likely to buy a product based on past data, but such a prediction is disconnected from an explanation of the deeper reasons—why those customers value the product or how their social or cultural context influences their decisions. This lack of transparency shows the core difference between prediction and understanding: GenAI identifies what happens without providing insight into why it happens.
Understanding (Verstehen) and GenAI Verstehen requires seeing the world through the eyes of others—understanding their motivations and experiences. For instance, a researcher studying leadership might explore how employees <i>feel</i> about their leader’s actions and what these actions mean to them. GenAI cannot properly empathize or interpret; it can only process patterns and generate outputs. Its black-box nature thus limits its ability to provide explanations.
Conclusion GenAI is useful for identifying patterns and making predictions, but, from the Verstehen perspective they cannot truly offer understanding. Understanding human behavior in its full depth—considering intentions, emotions, and context—remains a uniquely human task. The black-box nature of GenAI underscores this key limitation, which researchers must carefully consider.

Learning expectations

Overall, the learning expectation of these last three sections are that you should have obtained an understanding of the two different approaches (causality vs. understanding), what their aims are and what it means to provide a theoretical explanation in these approaches. This also implies having a sense of how there can be different levels of expectations for what it means to provide a theoretical explanation.

2.4. General reflections on theory

In this section a few more reflections on theories will be presented. The reflections are general and apply to both perspectives outlined above.

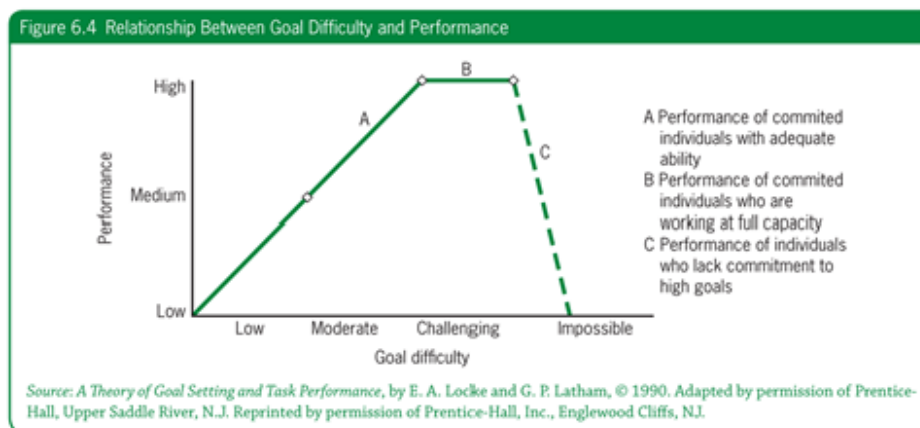
Structural vs. individual explanations

First, theories can be developed at different levels. A very typical and general distinction is between the structural or individual level. For example, we could try to investigate why some geographical regions have more entrepreneurs than others. This could be due to the individual characteristics of the people living in the region, like willingness to take risks. Or it could be due to structural factors, like geography or regional policies. Both explanations could of course be true and complementary, but they could also be competing. Some structural reforms might also be implemented in order to change individual risk willingness, and the

levels of explanations are likely to be connected. When pursuing theoretical explanations (e.g. in a student project) we should not necessarily investigate all possible levels and perspective. Rather, we should try to identify and pick promising avenues, which might entail to focus on one or the other level in a given situation.

Theories vs. models

While theories and models are sometimes used interchangeably (e.g. in Whetten 1989), they are different terms and typically refer to different usages. Briefly put, you can expect the following terminology to be relied upon: A model is not the equivalent of a theory. Yet, just as descriptions involve a theoretical basis, so do models. Usually, models constitute a concrete manifestation and visualization of theoretical relations. More precisely: *"I understand theories as bodies of knowledge that are broad in scope and aim to explain robust phenomena. **Models, on the other hand, are instantiations of theories, narrower in scope and often more concrete, commonly applied to a particular aspect of a given theory, providing a more local description or understanding of a phenomenon. Evolution and gravitation are theories, but their application to mate selection and the motion of the planets is best done via models. From this perspective, models serve as intermediaries between theories and the real world.**"* (Fried 2021: p. 336, my emphasis). The term model can refer to a statistical model (e.g. a linear regression) which are a means to represent data. The term can also more generically refer to a visual representation of the relevant factors and how they relate, as illustrated in the model inserted below. The strength of such a model rests in the power of visualization, which can give the reader a more immediate overview of the relevant factors and how they are related. That is, the model presents the main variables and also how the different variables are related - but not why they are related in this way.



Cited from Sinding et al. 2018: p. 191

Such a model would usually be an integrated part of a theory, where we then also would explain why goal difficulty and performance are related (an explanation you in fact will find in the text adjacent to the model). I will in a later section address the difference between a theory and hypothesis.

2.5. Student projects and theory

In the appendix (see end of this document) I've written about how and why theories are relevant for student projects, and how to search for theories, e.g. for your bachelor project.

3.0. How to we to get a theoretical explanation?

"Facts do not simply lie around waiting to be picked up. Facts must be carved out of the continuous web of ongoing reality, must be observed within a specific frame of reference..." (quote from the cover of Rasmussen et al. 2006).

In the prior sections we have addressed how theories are defined, and what an explanation actually is, according to various perspectives. In this section I will present insights on how we get to an explanation, either in the sense of validating it via *testing* the theory or *building* it in the first place.

3.1. Theory testing

A common way of distinguishing between theory testing and theory building is to categorize theory testing as a deductive process, and theory building as an inductive process. I will present an addition to this simplified characterization in a later section (3.3.1), yet for now merely present it. In the figure below theory building and theory testing are visually related to inductive and deductive reasoning processes.

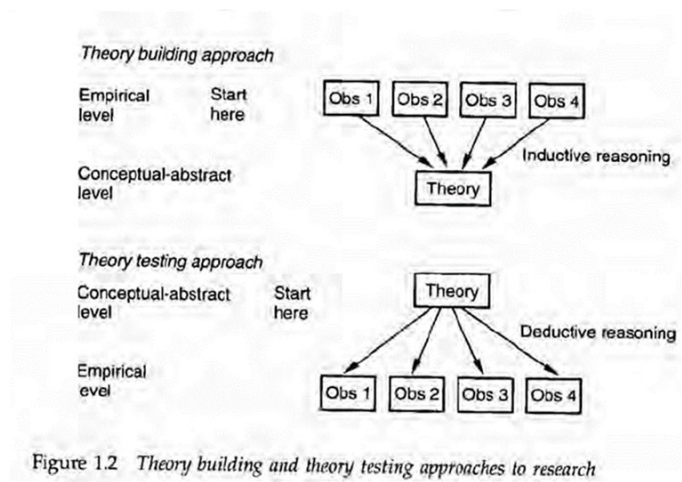


Figure 1.2 Theory building and theory testing approaches to research

Source: De Vaus 2001, p: 3-4

As the figure shows, in an **inductive** reasoning process, you start with observations and use them to **build theory**. After making observations, you might inductively be able to see a pattern emerge – based on these patterns one can then form a theoretical explanation. In contrast, in a **theory testing** approach, you already have a theory in place to **deductively** test before you begin to collect empirical data (that is observations). De Vaus (2001: p. 8) notes: *"If the theory is true certain things should follow in the real world"*, in other words certain observations should be deductively implied by the theory. Existing theory is thus used to inform how to collect data and what results to expect (e.g. by making hypotheses). **Hypotheses** are in fact predictions that are deductively derived from the theory, and not inferred from the data. For example, if you believe that incentives lead to better performance, then you can deduce the hypothesis that higher monetary rewards (a specific kind of incentives) should improve performance. Ideally, the hypotheses should therefore be finalized before even looking at the data. A researcher committed to a deductive approach should thus not develop a theoretical expectation, collect data, find a different pattern than expected, report this different pattern and then still present it as a deductive finding. Do note that while deductive research **ideally** starts with clearly specified hypotheses and analysis plans, the **reality of research often requires making some analytical decisions after data collection begins**. This isn't necessarily a flaw in the deductive approach, but rather a normal part of the research process. The key is to

be **transparent** about which decisions were pre-specified and which were made based on the data, and to consider how these decisions might affect the strength of conclusions that can be drawn. Overall, this is also why visual presentations of the **deductive research process are sequential and strictly linear**, going from theory to hypothesis to data collection to findings and so forth (see e.g. figure 2.1 in Bell et al. 2022: p. 21, Chapter on Business Research Strategies).

In order to document that a result was in fact deductive (and not inferred after looking at the data) you could ideally formally pre-register your theoretical expectation. That is, you could post the hypotheses and expected analytical approach on a platform, before collecting data – usually such pre-registrations are hidden during the data collection period, in order not to influence potential participants. Yet, as I will address in a later section, it is not always easy to predict all the decisions you have to make in a research study nor all the results you obtain.

Open vs. closed approaches

Another illustrative way of distinguishing between theory testing and theory building is by considering the terms '**closed**' and '**open**'. In theory testing, we engage in a 'closed approach' because we have a specific theory in mind that we set out to test. One asks: "I believe this to be the case, but what is the evidence for (or against) it"? The researcher would not ask an open question about what might be the case, but rather a specific question: is this theory the case? In other words, the researcher makes a bet and promise by saying; exactly this is what I want to test, and if I find this, I should have stronger confidence in my theorizing.

In contrast, in an 'open approach' you look at data and report whatever patterns are found without making any specific predictions or hypotheses beforehand (more on this in section 3.2). The distinction is important, because if a pattern is found inductively, there is a higher risk that the pattern identified (but not predicted) is spurious, i.e. just random and due to the fact that we too often find (random) pattern in the world (see this site which contains many fun and curious examples

<http://www.tylervigen.com/spurious-correlations>). To illustrate, you can throw a die 50 times and you might in attempts 49 to 50 get two sixes in a row. If you had stated before throwing the dice (i.e. pre-registered it), this might be an impressive prediction. Yet, just reporting this after the fact ignores that any 50 throws in a row with a die is bound to lead to some unusual patterns. Similarly, if you pre-specify that higher incentives will have a certain effect and you are right, this is more impressive than simply reporting it after the fact. To reiterate, this is why theory testing is considered a closed approach, since we are only testing the theory that was predetermined before any data was collected, rather than openly analyzing the patterns that emerge after data has been collected (which would be induction, more on this in section 3.2).

3.1.1. Theory vs. hypothesis (and models)

What is a hypothesis and how is it different from a theory?

"...in contrast [to a theory] a hypothesis is more specific yet also more speculative: a hypothesis makes a precise prediction about what should happen, in particular conditions, if the underlying theory is an accurate representation and explanation of the phenomena in question - predictions that are testable through observation (i.e. the collection of specific data)." (Gill & Johnson 2010: p. 43).

A **theory provides a fairly general explanation to (why) questions** (as outlined in section 1.3). In contrast, a **hypothesis is a specific, testable prediction that is narrower** and focuses on a specific manifestation of a

theory. A hypothesis is derived from a (more general) theory and is used to test the theory through the collection of specific data. The word 'derive' can be difficult to interpret here. Ideally, a hypothesis follows logically from the theory. In practice, it is not a deduction like 1) all men are mortal, 2) Socrates is a man, therefore 3) Socrates is mortal. Rather, a hypothesis is a well-informed guess, based on known theory or possibly combining different theories. The hypothesis serves as a bridge between the theory and the data by specifying the theoretical expectations in a specific situation.

A hypothesis is needed to bridge the gap between theory and data because a theory alone is too broad to be tested directly. For example, a theory might specify how and why incentives impact performance, but a hypothesis would specify exactly what is meant by 'incentives' and how performance is being measured in a given situation. For one, incentives can be determined based on the average performance, yet could also be based on the best possible performance (average sale, or the biggest sale). Second, incentives could be an increase in the monthly pay, but could also refer to bonuses of various sorts. Some of these specifications can be made explicit in a hypothesis (e.g. by writing "nr. of sales executed solved" to indicate how performance is measured). **A hypothesis is thus connected to the kind of data is to be collected.**

Note, that just because a hypothesis might not be supported (p-value over 0.05), does not mean that the hypothesis is false. **Absence of evidence does not mean evidence of absence.** Think of the die example: Assume a die is slightly uneven, leading to 0.5% bigger chance of a 6, than it should. Throwing the die 100 times is not going to lead to an unusual amount of 6s, in a statistical significance sense. Yet, **just because no evidence was found, does not mean the evidence cannot be there.**

A hypothesis could deductively extend an already established theory, or a hypothesis could also be a complete match to an already conducted study. The value in the latter situation could be that the sample is different, or the industry is different, or that the study will be carried out in a different time. In this situation the theory is tested again, ensuring (or not) robustness even in a new, previously untried, situation. If the theory is not validated, this might mean that the theory should be more narrowly specified or that it is wrong. The value in the former situation where theory has been extended, is that the theory will be further strengthened in scope, expanding and specifying the theory.

3.1.2. If a theory is supported, is it true?

If you have found support for the hypothesized effects, you have found support for your theory. Yet, does this mean the theory is true? Earlier in the 20th century it was a widespread belief that if you found that your data fitted your theory, then there were reasons to believe the theory was true (see Kuhn 1970).

Let us assume you find evidence for your hypothesis and therefore theory. You might even have conducted several studies that all find support for it, relying on different operationalisations, that is measurements, tasks, contexts, samples etc. Yet, we cannot rule out that the next study will not find support or might even find an opposite effect. In that sense, it is difficult if not impossible to be absolutely certain that all future studies will lead to the same result, still supporting the theory. For example, in economics most researchers for a long time thought that raising the minimum wage would reduce the supply of jobs (cf. theory on supply and demand), even to the degree that the Nobel prize winner Buchanan stated that the result is as certain as water not running uphill (see quote in Smith 2021). However, later more careful and precise studies have shown that there probably is no strong relation between the minimum wage and number of jobs (Smith 2021). Note that even if you find support in all studies conducted, in economics and business administration the results would still vary; maybe sometimes finding a strong effect and sometimes only a

weak effect (see any typical meta-analysis, that synthesises results from many studies, Harrer 2021, curriculum in a later lesson). You can contrast this with some areas in physics, where every single (correctly done) study will find the exact same result. Or, more precisely, a discrepancy might be well defined and appear numerically small. A recent example showed that a known physics result might have to be updated by 0.00000000502 (Karlis and Spencer 2021). Confidently stating that a theory is true per se, is thus difficult, in most fields, at least in economics and business administration (Nelson 2016, curriculum in a later lesson).

Overall, this is why it is often argued that we should say a theory has been **supported**, rather than **confirmed**, or that the theory is now true.

3.2. Theory building

As illustrated in the figure below (same figure as in section 3.1), theory building is related to inductive reasoning. Here we begin with a range of observations (or data, more generally), and inductively infer and build theory based on patterns identified and created in the data. "*...inductive reasoning entails making general inferences about a phenomenon through the observation of particular instances of the phenomenon*" (Johnson & Duberley 2000: p. 16). **In other words, you observe the world (i.e. collect or create data) and try to find patterns.** Theory building is therefore an **open** approach, since we don't set out to either find support (or not) for a theory. You might start with a **theoretical focus point**; I wonder how incentives shape behaviour? Or, you might think incentives positively impact performance yet you keep an open mind and do not set out to test a given theory and hypothesis. You could also simply stumble open something interesting while investigating something unrelated. Maybe you are interested in leadership and performance and then suddenly realize something about the nature of incentives. These examples are meant to highlight that we usually begin with a focus point, e.g. a research question, or an existing theory. In other words, research is usually not completely open-ended, without any aim (Eisenhardt 1989). Inductive research is most likely shaped by prior expectations, whether explicitly or just implicitly.

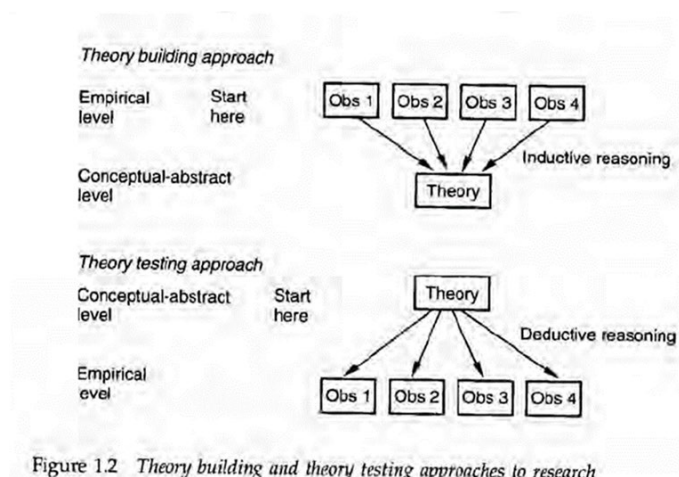


Figure 1.2 Theory building and theory testing approaches to research

Source: De Vaus 2001, p. 3-4

This also illustrates that pure induction occurs quite rarely (cf. discussion on grounded approaches, Bell et al. 2022: p. 532-538, Chapter on Qualitative Data Analysis), since we usually rely on prior theory. Thus, before data collection an inductive researcher would read the relevant theory, reflect on how others have designed their research in general and questions in particular. However, the aim is not to test a specific

relation (which would be deduction). The researcher might think that X and Y are related somehow, but the aim is not to establish if the existing theory can be supported. Rather, the aim is to develop further, potentially new insights.

Combining induction and deduction

Even if you have established a very convincing pattern and theory based on an inductive, theory building approach, it is difficult to be certain that what you have found is really true and would also be found in a different context, either today or in the future. The most famous example of this is the idea that all swans are white. All swans encountered in Europe were white – then someone identified a black swan in Australia. We don't know if the patterns we have identified generalize beyond the context where we have found them (yet see Flyvbjerg 2006 on how to generalize from case studies). Generally the aim of research is to find results that do not just hold true in one particular and unique case context, but can generalize to some degree. In order to ensure that the pattern can generalize, you (or other researchers) would then try to test the theories.

The traditional approach to theory testing and building involves first collecting data inductively to identify patterns and develop new or existing theories, followed by deductively testing these patterns (Bell et al. 2022). The process can also be iterative and non-linear, with multiple cycles of theory development and testing (see figure 27.2 in Bell et al. 2022). For example, you could begin with a deductive approach, and upon identifying unexpected findings, engage in further induction to make sense of what you have found. Finally, keep in mind that most terminologies and frameworks involve exceptions. For example, qualitative work can be deductive if it is structured, based on pre-defined constructs and with the aim to support or falsify something (Hyde 2000). For a discussion on whether one should prefer single or multiple case studies, see the discussion between Eisenhardt 1991, and Dyer & Wilkins 1991).

3.3. Induction, deduction or a mix?

The above basic distinction between induction and deduction, and between theory building and testing as two separate kinds of research processes, should immediately be followed by a substantial caveat.

"However, just as deduction often entails an element of induction, the inductive process is likely to involve some deduction" (Bell et al. 2022: p. 24, chapter on Business Research Strategies)

"Deductive and inductive strategies are better thought of as tendencies rather than as a hard-and-fast distinction" (Bell et al. 2022: p. 24)

As the quotes show, in practice the processes are often intertwined. In an inductive study, the patterns you find can be partially inspired or even driven by prior theoretical insights. Or, you might have started out with a pure inductive approach, but midway through data analysis, you might begin to seek saturation and ensure that the patterns you identify in the latter half, matches what you identified in the first half. This latter approach would be more akin to deductive reasoning (see also Hyde 2000). Similarly, in a deductive study, typically not all analytical decisions (about how to statistically analyze data e.g.) are made beforehand. Hence, some decisions might be driven by what is learned during the data collection or analysis process.

"...what is involved is not a crossroads where [you] go left or right. A better analogy is a complex maze where we are repeatedly faced with decisions and where paths wind back on one another" (Hammersley 1992, 172, cited from Johnson & Duberley 2000: p. 59).

3.3.1. Abduction

The following quote is from Heckman, an economist who won the Nobel Memorial Prize in Economics in 2000: *"Abduction is the process of generating and choosing models, hypotheses and data analyzed in response to surprising findings. All good empirical economists abduct. Explanations usually evolve as studies evolve."* (Heckman and Singer 2017: p. 298)

A range of different definitions and approaches to abduction exist. They generally highlight abduction as **'inference to the best explanation'**, thus **neither induction nor deduction**, but an attempt to find the best possible explanation. The following paragraphs are based on a fairly simple, useful conceptualization presented in the most prestigious journal in economics by a star scientist, and thus a good starting point. The main message of this account is that induction and deductions are 'ideal' reasoning processes, and that papers are often written up as if they were purely inductive or deductive - while in practice a mix is fairly common (van Maanen et al. 2007). It is important to note, that abductive reasoning processes according to this account do not just integrate elements from both induction and deduction but also refer to a **different kind of reasoning process**. This conceptualization is thus based on the premise that the pure inductive or deductive approaches are rare.

Ideally we inductively create patterns and then deductively test them. But often, you have a theory, test it, find it is missing features (an anomaly, a result we can't really explain and understand). This might lead a researcher to iteratively (and abductively) rethink the problem, while looking at existing theories and maybe even collecting new data in order to re-frame the theory, and potentially test it (again). In this process the researcher is not just switching between induction and deduction, but engaging in one overall research process that combines different kinds of thinking. The researcher both tries to find new patterns, while also testing the patterns. The end aim of abduction is the best possible explanation, not truth as such.

Abduction is thus used and appropriate when existing explanations of data and the world are not satisfying. You do not have to create new data or make new deductions to test a new theory, but can go back and forth between data and theory to rethink existing theory (van Maanen et al. 2007). Again, ideally all theories are fully developed prior to testing, yet in practice this does not always happen. According to Heckman and Singer (2017) and this take on abduction, the field of economics and business administration should more clearly acknowledge that we rarely make a clean deductive test. For example, every time we modify a hypothesis after having looked at the data, or every time we try out multiple statistical analysis procedures (after having looked at the results of the first attempt), you could argue that some kind of non-deductive reasoning is in play.

To illustrate, let us imagine I have data on incentives and performance from a company's archive. I might deductively have set out to test a certain relation, yet while investigating the data I identify patterns I had not expected. I might talk to individuals at the company to help me make sense of the data, and then proceed to develop new descriptions and explanations of the identified relations. I might then switch the theoretical framework and implement a new statistical model, in order to test new theoretical expectations and hypotheses. There is nothing wrong with this approach per se, as long as the results subsequently are not presented as being based on a pure, traditional deductive process, claiming that all theory was developed before looking at the data. Theory was built as well as tested on the same dataset.

A final note on the words induction, deduction and abduction

The following snippet is from <https://www.merriam-webster.com/words-at-play/deduction-vs-induction-vs-abduction>: *"If you have trouble differentiating deduction, induction, and abduction, thinking about their roots might help. All three words are based on Latin ducere, meaning "to lead". The prefix de- means "from", and deduction derives from generally accepted statements or facts. The prefix in- means "to" or "toward", and induction leads you to a generalization. The prefix ab- means "away", and you take away the best explanation in abduction."*

3.4. Unambiguous consensus on frameworks and terminologies?

There is no single, universally agreed-upon framework for thinking about the concepts discussed in these sections. It is important to recognize that many research projects do not fit easily into the categories of theory testing or building, or deductive or inductive reasoning. Furthermore, most textbooks equate inductive reasoning with qualitative approaches, and deduction with quantitative. Yet, as noted in the above, these are general rules of thumb and exceptions exist. We therefore should not think of these distinctions as too clear-cut (Heckman and Singer, 2017; Bell et al. 2022). Nevertheless, it can be helpful to reflect on the primary aim of your research project early on and decide whether you are primarily testing a pre-existing theory or seeking to understand an empirical context that is not well-explained by existing theories. This will provide a useful starting point for your research and make it easier to communicate your approach to others."

4. Next issue: How theories improve

Now that you have learned about various strategies for building and testing theories, including inductive-deductive and abductive approaches, you may be wondering how theories actually improve over time. It is often assumed that science progresses in a rational, accumulative manner, with the best evidence and data leading to stronger theories that build upon and surpass previous ones. However, it is worth considering whether this is always the case in practice. In upcoming lessons, we will delve into questions such as:

- Does scientific progress always happen in a rational and accumulative way?
- Should we consider well-tested theories to be true, almost true, or simply supported by multiple observers? Or should we rather than aiming to find out if theories are true, try to find out if theories are false?
- Do these theories represent and directly mirror external realities in the world, or are they constructed by researchers?

Appendix: Theory and student projects (maybe useful for you)

In a large student project (e.g. for a bachelor project), students are expected to "*systematically identify and choose relevant theoretical literature*" and apply theories (see e.g. this course description at HA/BSchb (Aarhus University): <https://kursuskatalog.au.dk/da/course/116133/Bachelors-Project>). How do you do this, and why should you? Theories are helpful because you want to build on the insights and knowledge of others rather than starting from scratch or repeating mistakes that have already been addressed. By searching for relevant theoretical literature, you can learn how others have conceptualized similar research challenges, what research designs and methods they have used, what concepts are relevant, and what results you can expect. Textbooks or the curriculum of a given university course are not likely to provide all of the information you need. Just the number of ECTS points in a typical bachelor project (20 or 30) can make you realize, you should go deeper or broader than the literature provided in a (e.g.) 10 ECTS points course.

Yet, how do you then search the theoretical literature? I list some of the key steps in the following, and you can also see a more extensive description here (<https://paperpile.com/g/google-scholar-guide/>).

- 1) Identify what your key search terms should be (maybe just 2-3, maybe a bit more). Think carefully about what the important words to use should be, since a good search can save you a lot of time. Note that advanced search options usually allow for efficient search processes, e.g. enabling combining words or allowing inflections of words.
- 2) Go to your library website or Google Scholar and search.
- 3) If you are getting too many (hundreds), or too few hits (zero, or just a few), then adjust the use of key words.
- 4) Go through the results by looking at the titles and the abstracts to identify work you think might be useful. Do also pay attention to how recent studies are and how often they are cited by others in subsequent studies. In Google Scholar this latter citation number is found just below the information of the text in, cf. the example below.

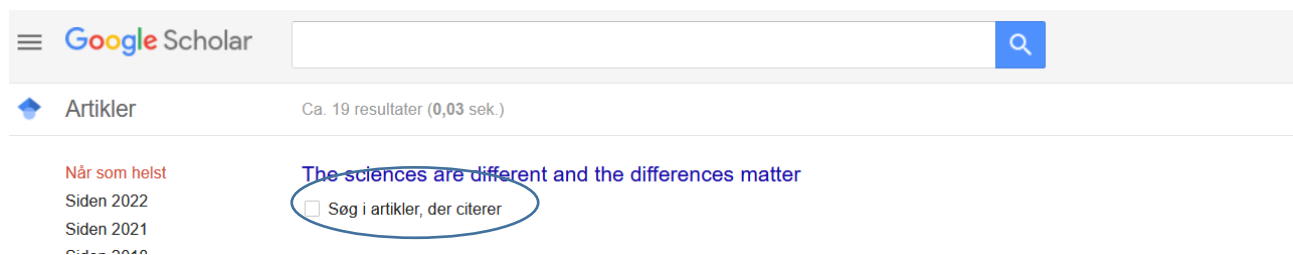
The screenshot shows a Google Scholar search interface. The search bar contains the text "nelson 2016 the sciences are different". Below the search bar, there is a list of results. The first result is titled "[HTML] The sciences are different and the differences matter" by RR Nelson, published in Research Policy, 2016, Elsevier. To the right of the title is a link to the full text: "[HTML] sciencedirect.com". Below the title and author information, there is a brief abstract. At the bottom of the result, there is a citation count: "Citeret af 19", which is circled in blue. Other links like "Gem", "Citer", "Relaterede artikler", and "Alle 4 versioner" are also visible.

Note: If there is a link to the right of the paper (as above, see upper right corner) then you freely and directly access the article even if you are not logged into a library system. Also, if you click on the link just to the left of the circle above ("cite"), then you get the bibliographical information you need to put into the reference list.

It is difficult to provide a general baseline for what 'many' citations are, but you can certainly compare the number of citations within your sample of search results, to figure out what is high or low number in your

setting. Note that Google Scholar already has filtered your search results based on popularity, just as it does in the 'regular' Google search. While popularity does not equal quality, it can be a useful first indicator in a student project.

5) When you have found relevant papers, you can also investigate which later papers have cited this original work, to pursue further angles. Do this by clicking the "cited by" link, cf. the circled text above. This can be a very efficient way to identify the most relevant prior studies. You can also search within that limited list of papers. This can be done by clicking on the "cited by" link and then tick "search in articles that cite" box (see below).



6) During the subsequent process of reading you are likely to encounter additional relevant terms, and continuing the search can provide further useful references.

In the above I have written that you should search for 'relevant' papers, but that in itself of course doesn't tell you what constitutes a relevant paper, for you. General guidelines can appear ambiguous yet can still be helpful. The popular book "The good paper" offers some guidelines on this topic (<https://samfundslitteratur.dk/bog/good-paper>). Some criteria to consider when evaluating the relevance of a paper include: i) how closely the paper's methods, concepts and empirical context align with phenomena you are studying ii) the clarity and consistency of the paper's explanation for the phenomena (is it a good paper?) iii) the empirical support for the theory and paper (again, is it a good paper?) iv) whether the paper provides complementary or conflicting perspectives in terms of methods, concepts, and results. Overall, these criteria could help you determine whether a paper is a good fit for your research.

Let me provide an example: Let us assume I am interested in how the size of a reward is linked to performance: I go to a search engine (e.g. Google Scholar) and begin with a fairly specific search: "size of reward related to performance". If this doesn't work out, I would try other concepts, such as incentives, money, salary etc. Ideally, I get results that use theoretical concepts that I am already familiar with, which can further help me identify texts that have a research design, theoretical framework and results that can assist my work. Just to repeat, the search is not merely supposed to give me information on what results to expect, but also help me make decisions on the quality of my original research question, what theories might be relevant to address my question, what research designs others have relied upon, what specific decisions they have made when dealing with data and of course what results former studies have led to etc. When assessing the search results, I would give the studies that seem the most relevant to my research question a quick (first) read, and ideally also investigate what other studies have cited these original studies, to ensure a broader search. This could take a couple of hours, if my question is very well-defined, yet could also easily take full days to get me to a set of papers including perspectives and a terminology that I find useful. It is generally a good idea to prioritize this part of the project, since time invested here can make the subsequent work easier.

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If you plan to cite this document, e.g. in a bachelor project, use this reference:

Bergenholtz, C. (2025f). What is a theory and what constitutes a theoretical explanation? Course material, Aarhus University Brightspace.

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